

Topic: Selection Operation

Q. Explain the Selection operation in Relational Algebra. Write its syntax and provide examples demonstrating how it filters tuples.

[DEF] Definition to Remember

The Selection Operation (denoted by sigma) is a unary operation in relational algebra that filters and returns specific rows (tuples) from a relation that satisfy a given logical condition (predicate). It is equivalent to the WHERE clause in SQL.

Selection Operation (sigma)

* Operates on a single table (unary).

2. Syntax of Selection

$\sigma_{\langle \text{predicate} \rangle}(\text{Relation_Name})$

* Predicate: Uses comparison operators (=, !=, <, <=, >, >=) and logical operators (AND AND, OR OR, NOT NOT).

3. Examples

Input Relation: EMPLOYEE

Emp_ID	Name	Department	Salary
1	Alice	IT	50000
2	Bob	HR	40000
3	Charlie	IT	60000
4	David	Sales	45000

Example 1: Simple Condition

Query:

Example 2: AND Condition (AND)

Query:

Example 3: OR Condition (OR)

Query:

[MUST-WRITE] Must-Write Points (for 10 marks)

- Selection is a unary operation -- operates on one relation.
- Denoted by sigma (Sigma); equivalent to SQL's WHERE clause.
- Filters rows (tuples) based on a logical predicate -- horizontal filtering.
- Output has the same columns as input but only matching rows.
- Predicates use comparison operators (=, !=, <, >) and logical operators (, , ¬).
- Can be nested: apply selection to the result of another operation.
- Automatically eliminates duplicate rows (set semantics).

[RECALL] Quick Recall

sigma (Sigma) -> Unary -> Horizontal Filtering (rows) -> Predicate (=, !=, ,) -> Equivalent to SQL WHERE -> Can be nested